Scioly Summer Study Session - fuzzylogic - Repaired practice edition - Answer key

Division C | Astronomy | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

1-a. (4 points) Authored replacement for source item Question 1-a: Why not model a galaxy as one star?

Unit: Galaxies, stellar populations, and the interstellar medium.
Worked-case data: A region is dark in visible light but bright in infrared and contains embedded young sources. Explain without declaring it empty.

answer: Integrated light combines many stellar temperatures, evolutionary stages, gas, and dust with differing attenuation and emission. (4 points).

1-b. (6 points) Authored replacement for source item Question 1-b: Solve this case and explain your reasoning: A star’s luminosity is fixed. Its distance doubles. Predict its observed flux and name an independent check.

reasoning: Use F = L/(4πd^2).
Replacing d by 2d multiplies the denominator by 4.
The new flux is F/4; substitute it back to recover the same luminosity. (3 points).

conclusion: A distance change affects apparent flux, not the star’s intrinsic luminosity. (3 points).